human annotators contracted by Stiennon et al. (2022) to choose summaries according to a list of criteria designed to select high quality summaries.

**Instruction Following** (Chung et al., 2022; Dubois et al., 2023; Wang et al., 2022) is one of the main use cases for the LLM fine-tuning techniques we investigate, so we also evaluate models trained in this setting.

We use the SFT, RLHF, and RM models released by Dubois et al. (2023, AlpacaFarm). These models were trained in a very similar way to how we train our summarisation models, and all based on the LLaMa 7B base model. Models take a text instruction as input and are trained to output preferred answers to those instructions. The outputs used for the SFT model are from the `text-davinci-003` model from the OpenAI API, and the human preferences used to train the reward model are gathered by the authors of AlpacaFarm based on outputs of the SFT model. For precise details on how these models were trained, refer to Dubois et al. (2023).

## 5 Model Evaluation

We now describe how we evaluate both out-of-distribution generalisation and output diversity for the different fine-tuning techniques.

### 5.1 Generalisation Evaluation

To evaluate the performance of our trained models, we use GPT-4 (OpenAI, 2023) through the OpenAI API [1] as a *simulated human annotator*. While we ultimately care about human preferences, these are expensive, difficult to gather, and often noisy. Previous work has shown that GPT-4 accurately simulates human preferences in both summarisation (Rafailov et al., 2023) and instruction following (Dubois et al., 2023; Zheng et al., 2023), so we use it as the main performance metric in both tasks.

To evaluate a model's performance on a given dataset of inputs and reference outputs with GPT-4, we prompt GPT-4 with the input, the reference output, and the model output, and prompt it to decide which output is better. We use a variant of the prompts from (Rafailov et al., 2023) for summarisation, and alpaca_eval (Li et al., 2023) with the standard prompts for instruction-following. See Appendix D for the precise prompts and other details. This gives us a *percentage win rate of the model being evaluated versus the human-annotated reference output*, which we refer to as preference vs reference (**PvR**). In Appendix D.1 we validate that this evaluation is a good proxy for human preferences. We also perform *head-to-head comparisons between two policies*, by prompting GPT-4 in the same way to decide which of two model model outputs is better.

To evaluate out-of-distribution (OOD) generalisation, we specify an in-distribution (ID) test set and one or more out-of-distribution (OOD) test sets for each task, which have inputs drawn from a different distribution. In each of these sets we have evaluation inputs and corresponding reference outputs (produced either by humans in summarisation or `text-davinci-003` in instruction-following).

**For summarisation**, *the ID test set is the original TL;DR* test set from (Stiennon et al., 2022), and *the OOD test set is the CNN/DailyMail* test set, a dataset of news articles and corresponding summaries (Nallapati et al., 2016). This tests the ability of the model to have learnt the more general skill of summarisation and to apply it in a very different domain.

**For instruction following**, the *ID test set is a new test set generated in the same way as the training set was for AlpacaFarm, using the AlpacaFarm variant of Self-Instruct* (Dubois et al., 2023; Wang et al., 2023). Regenerating the test set ensures that it was not seen during training or model selection for AlpacaFarm models. For *the first OOD test sets, we use the AlpacaEval evaluation test set proposed in the original paper*. This is a set of inputs taken from a variety of open-source instruction following and dialogue training and evaluation datasets (Bai et al., 2022; Ganguli et al., 2022; Gudibande, 2023; Köpf et al., 2023; Wang et al., 2023; Zheng et al., 2023), curated by Dubois et al. (2023). For an *additional OOD test set, we generate a set of Sequential Instructions using an adjusted Self-Instruct protocol*. These instructions contain multiple steps in a single input, often building on each other, and require the model to ensure that they complete all the steps to produce a satisfactory outcome. For more details on this dataset see Appendix H.

[1] `https://platform.openai.com/docs/guides/gpt`

We also report the *generalisation gap* (the difference between in-distribution and out-of-distribution performance) which provides a measure of generalisation specifically. Lower generalisation gaps imply the model generalises better, as model performance does not drop as much when the model is evaluated out of distribution. For the head-to-head comparisons between models, we look at the change in head-to-head winrate to give different evaluation of generalisation; the model whose winrate increases generalises better.

### 5.2 Diversity Evaluation

To evaluate the output diversity of trained policies, we use several diversity measures which are well-supported by prior work, namely **distinct N-grams** (Li et al., 2016), **Sentence-BERT embedding cosine similarity** (Reimers & Gurevych, 2019) and **NLI diversity** (Stasaski & Hearst, 2022). All of these metrics have been shown to align well with human diversity evaluations and with underlying diversity generators by Tevet & Berant (2021).

Each diversity metric $D$ takes a set of model outputs $O$, and produces a scalar score representing how diverse the set is. *Distinct N-grams* counts the number of distinct N-grams (averaging over $n = 1 \ldots_5$) in the set of outputs, and following (Liu et al., 2022) we use the expectation-adjusted distinct N-grams (EAD) formula to remove the bias towards shorter outputs. The *Sentence-BERT* metric embeds each element of the output set using a sentence transformer, and then measures the average cosine similarity between the embeddings. The metric is then 1 minus the average similarity.

The *NLI diversity* metric measures the number of entailments and contradictions between pairs of elements in the output set using a natural language inference (NLI) model and rates an output set as more diverse if it produces more contradictions and fewer entailments. We pass sentences sampled from elements of the output set (rather than complete elements) to the NLI model so that the inputs are closer to the model's training distribution.

For each policy $\pi$ we produce a set of $K$ outputs from the model for each of a subset of $N$ inputs from the test set, sampling with temperature 1:

$$\text{for } i = 1 \ldots N : \mathcal{O}^i_\pi := \{y_j \sim \pi(y|x_i) | j = 1 \ldots K\}.$$

In this work we use $K = 16$, $N = 500$. For a diversity metric $D$ we then evaluate the **per-input diversity** which is defined as the average diversity of the output sets over inputs (i.e. the diversity of $\pi(y|x)$), as well as **across-input diversity** defined as the diversity of outputs across inputs (i.e. the diversity of $\pi(y)$):

$$\text{PerInputDiversity}_D(\pi) := \frac{1}{N} \sum_{i=1}^{N} D(\mathcal{O}^i_\pi) \quad (2)$$

$$\text{CrossInputDiversity}_D(\pi) := D\left(\bigcup_{i=1}^{N} \mathcal{O}^i_\pi[1]\right) \quad (3)$$

Here $\mathcal{O}^i_\pi[1]$ is the first element of the set $\mathcal{O}^i_\pi$. For conciseness, we refer to expectation-adjusted distinct N-grams, sentence-BERT average cosine similarity and NLI diversity as EAD, Sent BERT and NLI respectively. We can view them as measuring *syntactic, semantic and logical diversity*, and hence using all of them ensures that we are evaluating diversity in a wide range of ways.

## 6 Experimental Results

In this section we present the results of our experiments on generalisation and output diversity. In general we find that RLHF performs better than SFT in an absolute sense both ID and OOD, but generalisation gap and head-to-head metrics tell a more nuanced story. However, RLHF does reduce output diversity substantially in the per-input setting, and still reduces it to a lesser extent in the cross-input setting. These conclusions are supported by similar results for OPT models across a range of model scales in the summarisation task, presented in Appendix J.

### 6.1 Generalisation

**Summarisation.** Fig. 2 shows the *GPT4-PvR* for SFT, Bo16 and RLHF policies trained on the summarisation task, showing the ID (TL;DR) and OOD (CNN/DailyMail) performance and the

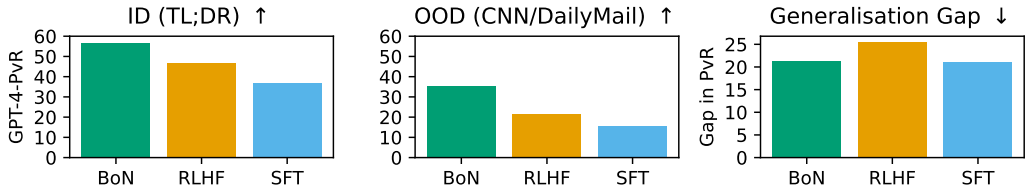


Figure 2: **Summarisation Generalisation Results.** GPT-4-PvR for SFT, BoN and RL policies, based on LLaMa 7B, trained on the summarisation task. In-distribution is performance on TL;DR, and out-of-distribution is on CNN/DailyMail, and generalisation gap is ID – OOD performance.

generalisation gap; Bo16 outperforms RLHF which outperforms SFT, both ID and OOD. While Bo16 outperforming RLHF is somewhat surprising, there are examples of this in the literature (Nakano et al., 2022), and Bo16 has a substantially higher inference-time cost, making RLHF better for practical applications. The performance of all policies drops OOD, which is unsurprising given the difficulty of the shift (reddit post summarisation to news article summarisation). The generalisation gap is fairly similar between methods, implying that none of these methods has a particular advantage with respect to generalisation specifically in this setting.

These results also show that the generalisation gap for SFT and Bo16 policies are the same (and they are still the same for a different choice of temperature for Bo16, see Fig. 9). This can be explained by the fact that the reward model generalises near-perfectly to the CNNDM preference dataset from (Stiennon et al., 2022). ID and OOD accuracy is 75.8 and 71.6 respectively, and considering that the maximum inter-annotator agreement in the CNNDM preference dataset is lower than TL;DR (and hence the maximum accuracy attainable is lower), it is plausible that the RM is not suffering any real drop in OOD performance in this case. This implies that all of the drop in performance for Bo16 is driven by the drop in performance of the SFT model, as if both SFT and RM performed worse OOD we would expect those drops in performance to compound. Overall, this shows that if you can expect your reward model to generalise well then BoN is a good choice of policy, although it is limited by the generalisation abilities of the underlying model being sampled from (the SFT model in this case), and is more expensive at inference time than RLHF and SFT.

In Appendix J we present results for a range of models based on OPT (Zhang et al., 2022), trained in a similar way to the LLaMa models but on split versions of the TL;DR dataset, and evaluated with a LLaMa proxy reward model. These results show similar trends: BoN outperforms RLHF which in turn outperforms SFT at the largest model size, and the ordering holds OOD for 3 different splits of the training dataset. This shows our results are robust across different evaluation metrics and base models.

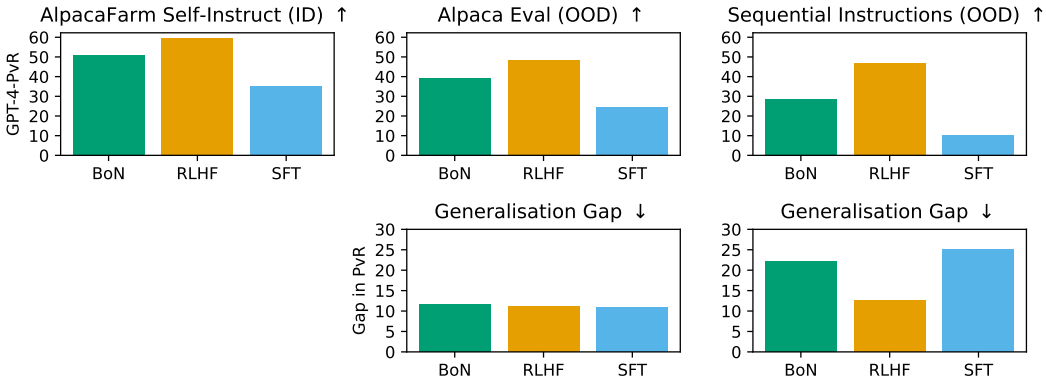


Figure 3: **Instruction Following Generalisation Results.** GPT-4 PvR for SFT, BoN and RL policies, based on LLaMa 7B, trained on the AlpacaFarm Self-Instruct instruction following task. ID is on AlpacaFarm Self-Instruct, OOD is on the AlpacaEval and Sequential Instructions datasets respectively, and generalisation gap is ID – OOD performance.

**Instruction Following.** Fig. 3 shows the results of BoN, SFT and RLHF models trained in AlpacaFarm Self-Instruct (Dubois et al., 2023) evaluated ID and OOD on AlpacaEval and Sequential Instructions. Similar to summarisation, we see that RLHF and Bo16 both outperform SFT, but here RLHF outperforms Bo16 across all datasets, in contrast to the summarisation task. As the focus of this paper is mostly comparing RLHF and SFT, we did not investigate this result further, but there could be many possible reasons for the change in ordering between RLHF and Bo16, as the two tasks are very different and the model training procedures are not identical.

We see that on AlpacaEval (the easier OOD generalisation task), models all generalise equally well, but on the harder Sequential Instructions OOD task, RLHF generalises much better. This suggests that *RLHF may generalise better relative to SFT for larger distribution shifts*, which potentially explains why models fine-tuned with RLHF have been observed to be much better in practice when interacting with users (Touvron et al., 2023b; Ouyang et al., 2022, *inter alia*): when users interact with these models the distribution shift is quite pronounced and hence many inputs are more OOD, and this is where RLHF model performance continues to be high.

While GPT-4 PvR is a useful metric, it does not show a difference in generalisation (as measured by generalisation gap) between SFT, RLHF and Bo16 models on the easier AlpacaEval dataset. This could be due to these models having similar generalisation properties, or be a deficiency of the metric. To investigate this further, we use can look at the GPT-4 Head-to-Head winrate of SFT vs Bo16 and vs RLHF, which is shown in Fig. 4. These results show that both RLHF and Bo16 winrates *improves* vs SFT by approximately 3.5% from ID to AlpacaEval OOD. This implies that RLHF and Bo16 generalise better than SFT even in this case, emphasising the need for a range of metrics when evaluating model generalisation.

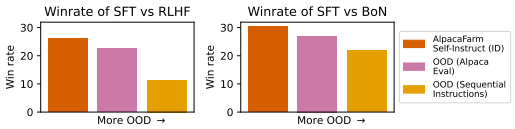


Figure 4: **GPT-4 Head-to-Head Winrate** of SFT vs RLHF and Bo16 in AlpacaFarm Self-Instruct, AlpacaEval and Sequential Instructions datasets.

### 6.2 Diversity

For the diversity evaluations, we focus on the summarisation task specifically, as it has the most compelling results. We ran some initial experiments evaluating diversity for the instruction-following models, but we did not see any meaningful differences. We hypothesise this is due to the diversity metrics we use being designed for settings where the model output is relatively short (e.g. a single sentence), whereas in the instruction-following setting outputs are generally much longer. Furthermore, RLHF models tend to produce longer outputs than other models, which can confound the evaluation of output diversity, since most metrics are not invariant to output length.

Fig. 5 shows the per-input diversity scores for RLHF and SFT models in the summarisation task. We see that across the first two metrics, RLHF has much lower output diversity than SFT. Fig. 6 shows the across-input diversity scores in the same setting. Here we see that while SFT generally has slightly higher diversity as before, the difference is much smaller than in the per-input case. The drop in across-input diversity cannot be explained purely by the use of the reward model, as BoN has similar or higher across-input diversity that SFT for the first two metrics. Both these trends are the same for OPT across model sizes (see Appendix J.4 for full results) We find that NLI does not show meaningful difference between models in either per-input or across input, showing that in a logical sense all models are similarly diverse.

The difference in across-input diversity between RLHF and SFT, while small, can also be taken as evidence of the phenomena of "mode collapse" hypothesised to occur under RLHF fine-tuning (janus, 2022). The hypothesised effect is that even for different inputs, RLHF models can be biased towards outputting text of a specific style or "mode", meaning that even changing the inputs to a model is not sufficient to generate truly diverse outputs. We believe that this is the first rigorous empirical demonstration of across-input mode collapse emerging from RLHF training specifically.

For most metrics and models, the across-input diversity scores are higher than the per-input diversity scores, which is expected given the across-input diversity distribution is much broader. However, for EAD (which measures diversity at the n-gram level), SFT has similar levels of per-input and